When Will a Borrower Default?

Survival Analysis in Credit Risk

Luke Munnelly

Should these be the same outcome?

- One borrower defaults after three months.

Hypothetical loans.

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- One borrower defaults after three months.
- Another defaults after two years.

Hypothetical loans.

Should these be the same outcome?

- One borrower defaults after three months.
- Another defaults after two years.

Both defaulted. The timing is different.

Hypothetical loans.

Principal still outstanding

Illustrative: €12,000 loan; €400 principal repaid each month.

After 3 payments

After 24 payments

Assume default just after payment; ignore interest, fees and recoveries.

Principal still outstanding

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After 3 payments	€10,800 outstanding
After 24 payments	€2,400 outstanding

Assume default just after payment; ignore interest, fees and recoveries.

Principal still outstanding

Illustrative: €12,000 loan; €400 principal repaid each month.

After 3 payments	€10,800 outstanding
After 24 payments	€2,400 outstanding

Earlier default can occur before much principal is repaid.

Assume default just after payment; ignore interest, fees and recoveries.

What do six months tell us?

Observed for six months; no default so far

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$$T_D > 6 \text{ months}$$

Provides information for the six months observed.

A binary label needs a horizon

- A classifier can predict default within 12 months.

Training and evaluation labels need appropriate follow-up.

A binary label needs a horizon

- A classifier can predict default within 12 months.
- Six months without default does not establish a 12-month label.

Training and evaluation labels need appropriate follow-up.

Which 12-month labels are known?

Observed history	Default by month 12?
Default at month 3	
Default at month 24	
No default through month 6	

Hypothetical loans; no earlier competing exit in these histories.

Which 12-month labels are known?

Observed history	Default by month 12?
Default at month 3	Yes
Default at month 24	No
No default through month 6	

Hypothetical loans; no earlier competing exit in these histories.

Which 12-month labels are known?

Observed history	Default by month 12?
Default at month 3	Yes
Default at month 24	No
No default through month 6	Unknown

Hypothetical loans; no earlier competing exit in these histories.

Why survival analysis?

- Use when events occur.

Machine-learning methods can also model survival outcomes.

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- Accommodate right-censored follow-up under suitable assumptions.

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- Use when events occur.
- Accommodate right-censored follow-up under suitable assumptions.
- Here: Cox for interpretable relative default hazards.

Machine-learning methods can also model survival outcomes.

Choose the event and the clock

Origination $\longrightarrow$ first recorded default

Elapsed time since the loan was issued

Follow-up can end in different ways

ILLUSTRATIVE · Hypothetical loans; months since origination



C: right censoring — active when observation ends.

D: closure — a competing event ending default exposure.

The survival function

First use a hypothetical default-only setting.

$$S(t) = P(T_D > t)$$

The survival function

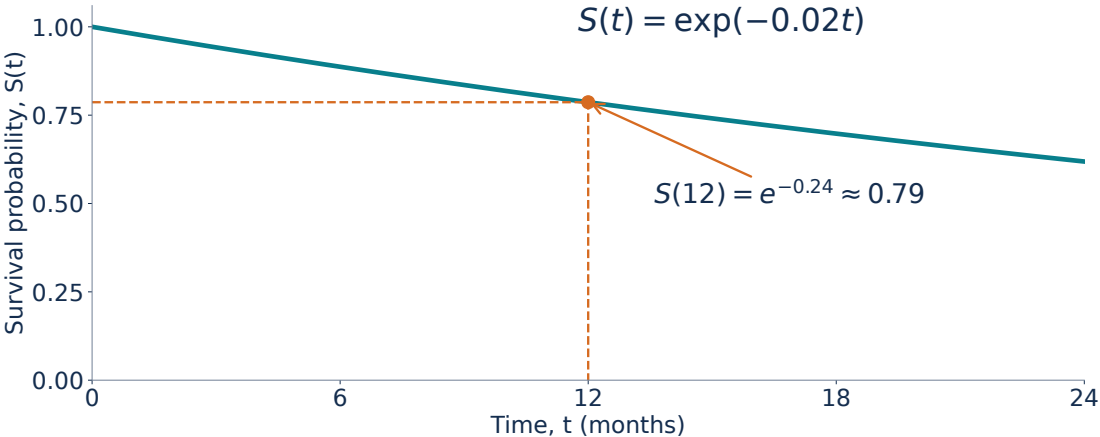
First use a hypothetical default-only setting.

$$S(t) = P(T_D > t)$$

The probability of remaining default-free beyond time t.

Survival: the probability of remaining default-free

Illustrative example · Default-only setting



At 12 months, about 79% remain default-free in this hypothetical setting.

How rapidly are defaults occurring now?

Among loans still exposed now:

how rapidly do defaults occur next?

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how rapidly do defaults occur next?

Hazard: an instantaneous rate, not a probability.

From a short interval to a hazard rate

Hypothetical default-only setting; t in months.

$$h(t) = \lim_{\Delta t \downarrow 0} \frac{P(t \leq T_D < t + \Delta t \mid T_D \geq t)}{\Delta t}$$

From a short interval to a hazard rate

Hypothetical default-only setting; t in months.

$$h(t) = \lim_{\Delta t \downarrow 0} \frac{P(t \le T_D < t + \Delta t \mid T_D \ge t)}{\Delta t}$$

$$P(t \le T_D < t + \Delta t \mid T_D \ge t) \;\approx\; h(t)\,\Delta t$$

Rate $\times$ a short interval $\approx$ a conditional probability.

From hazard to survival

Default-only setting.

$$H(t) = \int_0^t h(u)\,du$$

From hazard to survival

Default-only setting.

$$H(t) = \int_0^t h(u)\,du$$

$$S(t) = \exp\{-H(t)\}$$

With competing closure, default hazard alone does not determine actual default probabilities.

From timing to borrower characteristics

We have described event timing.

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How does the default rate differ between borrowers?

The Cox model

$$h(t \mid x) = \underbrace{h_0(t)}_{\text{baseline}} \underbrace{\exp(\beta^{\mathsf{T}} x)}_{\text{multiplier}}$$

For credit: a cause-specific default rate among loans still exposed.

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Baseline: the time pattern for reference covariates.

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The Cox model

$$h(t \mid x) = \underbrace{h_0(t)}_{\text{baseline}} \underbrace{\exp(\beta^{\mathsf{T}} x)}_{\text{multiplier}}$$

Baseline: the time pattern for reference covariates.

Multiplier: how the covariates scale that rate.

For credit: a cause-specific default rate among loans still exposed.

Compare otherwise identical borrowers

Hypothetical new vs existing customer; same baseline and other covariates.

$$\mathrm{HR} = \frac{h(t \mid \text{new})}{h(t \mid \text{existing})} = \exp(\beta_{\text{new}})$$

Proportional hazards: the ratio stays constant over time.

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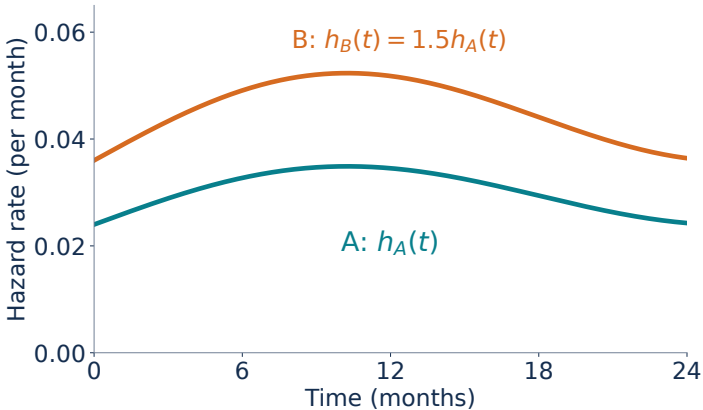
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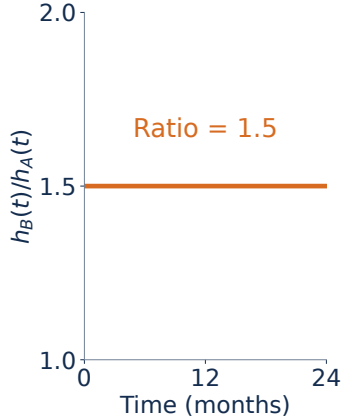
If $\mathrm{HR} = 1.5$: a 50% higher rate at each t.

Proportional hazards: the ratio stays constant over time.

Proportional hazards: changing rates, constant ratio

HYPOTHETICAL · A hazard is an instantaneous rate, not a probability





The baseline hazard can vary over time; proportionality concerns the ratio.

How does Cox learn relative effects?

Illustrative risk set: three borrowers just before one default.

A	B	C
Existing customer	Existing customer	New customer
Relative risk 1	Relative risk 1	Relative risk e^{β}

Which borrower accounts for this default?

Cox regression compares borrowers in the risk set

ILLUSTRATIVE · At risk immediately before a default; all other covariates equal

A	B	C
Existing customer	Existing customer	New customer
Relative risk: 1	Relative risk: 1	Relative risk: e^{β}

$$P(C \text{ defaults} \mid \text{one default in risk set}) = \frac{e^{\beta}}{2 + e^{\beta}}$$

If $\beta = \log(2)$: $\quad \frac{2}{1+1+2} = \frac{1}{2}$

Conditional on one default occurring now; not a probability over a time horizon.

The audited Bondora cohort

Estonia · Issued 2013–2015

Follow-up: at most 730 days from origination

Reporting cutoff: 20 July 2021; all loans have 730 days of potential follow-up.

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Estonia · Issued 2013–2015

Follow-up: at most 730 days from origination

9,844	6,924	2,508
Loans	Borrowers	Recorded defaults

Reporting cutoff: 20 July 2021; all loans have 730 days of potential follow-up.

Follow-up uses audited endpoints

- First recorded default date.

Closure is a proxy, not guaranteed full repayment.

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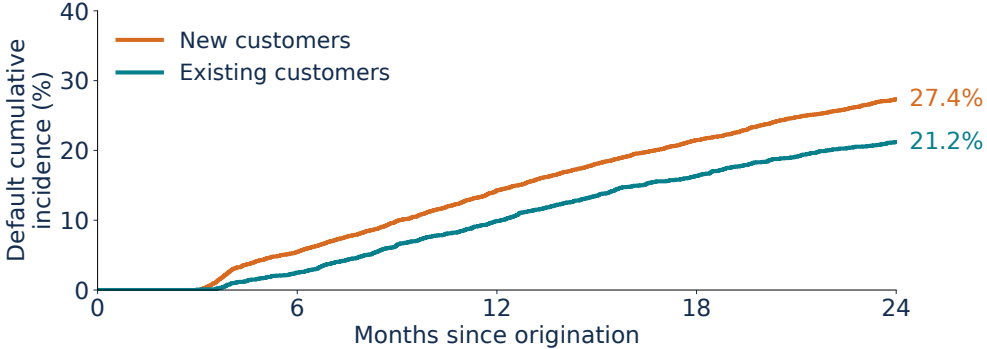
Follow-up uses audited endpoints

- First recorded default date.
- Recorded contract ending for loans labelled repaid.
- Follow-up ends at the first qualifying exit or 730 days.

Closure is a proxy, not guaranteed full repayment.

When do recorded defaults accumulate?

UNADJUSTED / DESCRIPTIVE · Closure proxy treated as a competing event



Loans at risk just before the indicated month

New	6,839	5,971	4,855	3,842	3,081
Existing	3,005	2,683	2,253	1,851	1,534

Aalen–Johansen · No uncertainty bands · Month = 730/24 days · Not a causal comparison

From timing curves to adjusted rates

Timing curves: unadjusted group differences.

Year-stratified baseline; borrower-clustered intervals.

Original monthly-schedule proxy; source currency/income period unverified.

From timing curves to adjusted rates

Timing curves: unadjusted group differences.

Cox: rates adjusted for these characteristics.

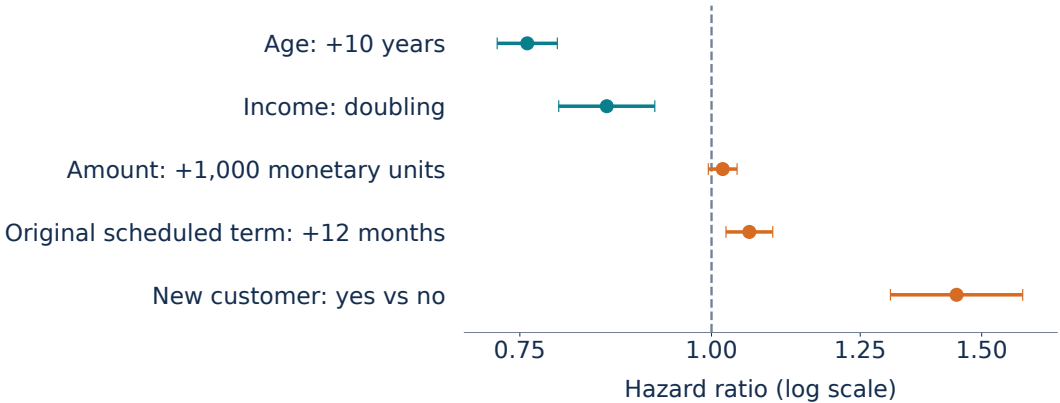
Age	+10 years
Income	Doubling
Amount	+1,000 recorded monetary units
Original scheduled term	+12 months
New-customer status	Yes vs no

Year-stratified baseline; borrower-clustered intervals.

Original monthly-schedule proxy; source currency/income period unverified.

Adjusted associations with the default rate

ADJUSTED · Audited cause-specific Cox model; origination-year baseline strata



Dots: hazard-ratio estimates · Bars: approximate borrower-clustered 95% confidence intervals.

Adjusted associations. Source monetary currency/income period unverified.

New versus existing customer

$$\text{HR} = 1.445$$

95% CI: 1.308–1.595

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95% CI: 1.308–1.595

About 44% higher conditional default rate among loans still exposed.

Association, not a default-probability ratio or causal effect.

Proportional hazards concern: this HR is a summary.

Where interpretation is limited

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- Proportional hazards concerns: age and new customers.
- Nonlinearity; approximate tied-date intervals.
- Recorded endpoints and measurement proxies.

Default risk changes over time

- Survival analysis uses both default timing and incomplete follow-up.

Group percentages are descriptive. Cox effects are adjusted associations.

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- In this cohort, about 27% of new-customer loans and 21% of existing-customer loans defaulted within two years.

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Default risk changes over time

- Survival analysis uses both default timing and incomplete follow-up.
- In this cohort, about 27% of new-customer loans and 21% of existing-customer loans defaulted within two years.
- Cox regression connects borrower characteristics to the default rate among loans still exposed.

Group percentages are descriptive. Cox effects are adjusted associations.

Sources

Bondora loan data — snapshot 20 July 2021, via Kaggle.

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Questions?